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SEMESTER 1 EXAMINATIONS 2025/2026

MODULE: CSC1048 - Computability and Complexity

PROGRAMME(S):

CASE	BSc Computer Apps (Sft.Eng.)
COMSCI	BSc Computer Science
ECGTE	Engineering & Computing GTE
ECSA	Study Abroad Program EC
ECSAO	Study Abroad Progrgam EC

YEAR OF STUDY: 1,3

EXAMINERS: Dr. David Sinclair (Internal) (Ext. 5510)

TIME ALLOWED: 2 hours

INSTRUCTIONS: Answer 4 Questions.
All questions carry equal marks.

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The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

Section A

QUESTION 1 ***[Total marks: 25]***

1(a) [5 Marks]

In the Haskell Programming language describe how *guards* operate with the use of example code.

1(b) [10 Marks]

Let a polynomial be represented as a list of coefficients. For example $a_3x^3 + a_2x^2 + a_1x^1 + a_0$ is represented as the list [a_0,a_1,a_2,a_3] . Define a Haskell function `evalPoly`, with the appropriate function type, which, given a polynomial and a value for x, will calculate the value of the polynomial for that value of x.

Hint: You may find the following identity helpful:
$a_nx^n + \ldots + a_2x^2 + a_1x + a_0 = a_0 + x(a_1 + x(a_2 + x(\ldots a_n)\ldots))$

1(c) [10 Marks]

Write a function `shortest`, with appropriate function type, that takes a list of lists and returns the shortest list in the list (and returns [ ] if the list of lists is empty).

[End Question 1]

QUESTION 2 ***[Total marks: 25]***

2(a) [5 Marks]

Describe function types in Haskell and the role that *class constraints* play in function types.

2(b) [10 Marks]

Write a Haskell program that takes a list of numbers (integer and floating point numbers) and returns the average of the non-negative numbers minus the average of the negative numbers. Consider 0 as a non-negative number.

2(c) [10 Marks]

Without using the built-in function `reverse`, write a Haskell function that tests if an array of elements is a palindrome, i.e. the reverse of the array is equal to the original array. Include its function type.

[End Question 2]

Section B

QUESTION 3 ***[Total marks: 25]***

3(a) [5 Marks]

Define *regular expressions*.

3(b) [10 Marks]

Formally define the machine that recognises languages generated by regular expressions.

3(c) [10 Marks]

State the *Pumping Lemma for Regular Languages* and what is its significance?

[End Question 3]

QUESTION 4 ***[Total marks: 25]***

4(a) [5 Marks]

Formally describe the Big-O Notation. Rank five typical Big-O categories from “quickest” to “slowest”.

4(b) [5 Marks]

In terms of Complexity, define NP-completeness.

4(c) [15 Marks]

The Cook-Levin Theorem states that *Satisfiability* is NP-complete. Give an outline of the proof of the Cook-Levin Theorem.

[End Question 4]

QUESTION 5 ***[Total marks: 25]***

5(a) [7 Marks]

Prove that the set $\mathcal{Q} = \{\frac{m}{n} | m, n \in \mathcal{N}\}$, the set of positive rational numbers is countable.

5(b) [12 Marks]

Prove that the *Halting Problem* is undecidable.

5(c) [6 Marks]

Give three other examples of undecidable problems.

[End Question 5]

[END OF EXAM]